AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 27 · STUDENT LESSON

Verify Important Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 23: show both a numerator and denominator when reporting a rate.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Separate an answer into claims and connect each claim to relevant evidence. Try the recall before opening notes.

Your learning target

I can separate an answer into **claim**s and connect each **claim** to relevant **evidence**.

I can recalculate quantities with consistent **unit**s and a reasonableness check.

I can report supported, contradicted and unresolved **claim**s without inventing missing **evidence**.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Prompts Are Design Instructions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Good Data, Bad Data. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Fluent AI **output** is not **evidence**; important **claim**s need **independent** support.



Your target

LEARNING TARGET A generated report says 18 of 24 tests passed, so success was **80%**. Verify the **claim** in two ways and write a corrected sentence.

Learner name:

Readiness for the connected explanation

A response follows the requested four-line format but invents a missing start time. Has it passed the task? Explain.

This week's end goal

By the end, I can verify quantities with consistent **unit**s and separate arithmetic from unsupported **claim**s.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Verify Important Outputs case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A generated paragraph confidently **claim**s that an animal lives in a region; the wording is not **evidence** that the **claim** is true.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

An answer can contain a mixture of correct, incorrect and unverified statements. Checking one sentence does not validate every other sentence. First separate the answer into individual **claim**s.

Then ask what **evidence** would support each one: a record, a measurement, a calculation or some combination.

Our fictional ribbon activity uses current materials card R. R states: four rolls of ribbon, each exactly 2 metres long; twelve pieces will each use 50 centimetres; no cutting loss is included in this paper model.

One metre equals 100 centimetres. An old card Q describes a different activity with three rolls. These are provided classroom records, not instructions to cut real materials.

A draft says “R gives four rolls. The starting length is 8 metres. The pieces use 600 metres. Two metres remain. Every participant will enjoy the activity.” Do not approve or reject the whole paragraph at once.

Inspect the roll count, starting length, used length, remaining length and enjoyment **claim** separately.

STUDENT LESSON · 3 OF 18

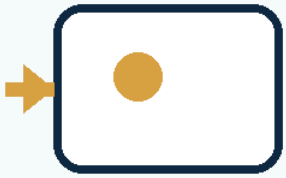
See how it works

> Follow the mechanism for Verify Important Outputs. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



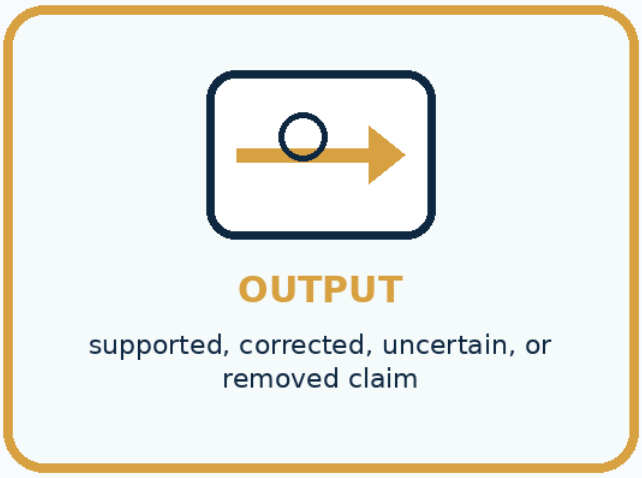
INPUT

specific claim extracted from an output



PROCESS

check independent suitable sources and reconcile differences



INPUT specific **claim** extracted from an **output**

PROCESS check **independent** suitable **source**s and reconcile differences

OUTPUT supported, corrected, uncertain, or removed **claim**

Key idea

CORE IDEA Fluent AI **output** is not **evidence**; important **claim**s need **independent** support.

How the mechanism works

UNDER THE HOOD **Verification** is **claim**-level **evidence** work. **Source** independence, authority, direct support, recency, and agreement matter. Multiple pages repeating the same upstream error are not **independent** confirmation.

Draw a fresh example with all three stages:

A closer explanation

Different claims need different checks

CLAIM	EVIDENCE PATH	DECISION
Four rolls	Read current card R	Supported
8 metres at start	4 rolls × 2 metres	Supported
600 metres used	12 × 50 cm = 600 cm	Wrong unit
Everyone will enjoy it	No enjoyment evidence	Unresolved

A correct source reference is a starting point. Read what the record actually says and check every derived quantity.

A **citation** is useful only if you can locate the **source** and confirm that it supports the specific **claim**. The title “Materials card R” beside a paragraph does not make the enjoyment **claim** part of R.

If a cited card Z is not supplied or cannot be found, that **citation** cannot currently verify its **claim**. Record the gap rather than inventing Z’s contents.

Check whether the **source** covers the right activity, version, population or time. Q’s three rolls may be accurate for its own activity but irrelevant to the current four-roll task.

In real work, also inspect who created the record, how they obtained the information and whether there is a correction or better **source**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

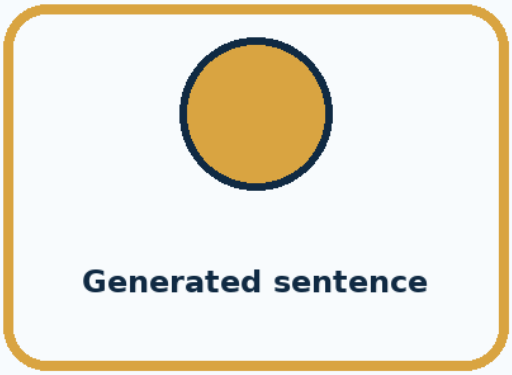
Compare the cases

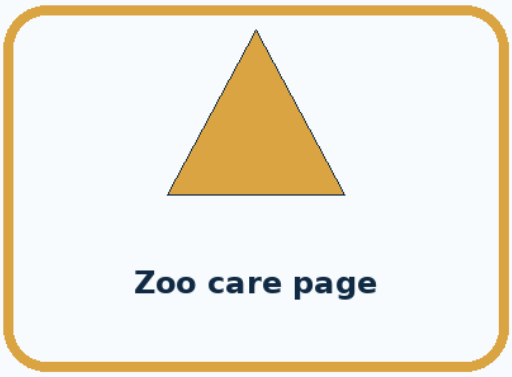
> Compare the supplied cases for Verify Important Outputs. Identify the detail that changes the conclusion.

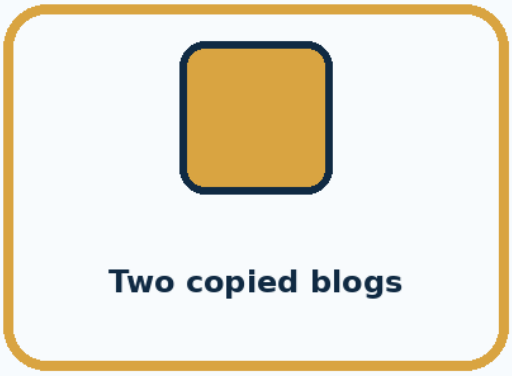
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Generated sentence Clue: fluent **output**. Meaning: **claim**, not **evidence**.

The case to investigate

CASE 2 — Zoo care page Clue: direct animal information. Meaning: possible **source**.

The case to investigate

CASE 3 — Two copied blogs Clue: same origin. Meaning: not **independent**.

The case to investigate

CASE 4 — Conflicting references Clue: **evidence** disagrees. Meaning: report uncertainty.

Keep this boundary clear

BOUNDARY Two search results are not two **independent source**s when both copy the same page.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Make ten bookmarks instead of eight.

Predicted result

Prediction: 100 cm, or 1 m, remains.

Why the result follows

Reason: 600−10×50=100 cm.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Verify Important Outputs to explain a concrete decision, rather than only memorising definitions.

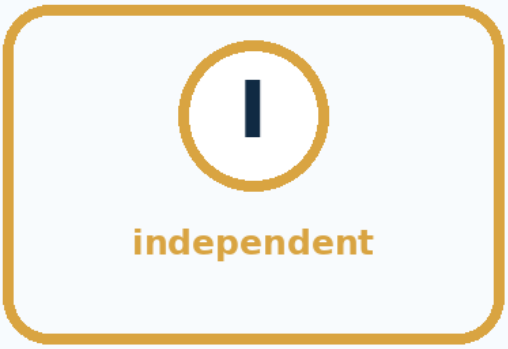
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
claim	statement that can be checked	
source	origin of evidence	

Word	Meaning	My example
independent	not merely copying the same origin	
corroborate	support a claim with separate evidence	

Misconception detective

> **Find the unsupported claim**
>
> **CLAIM** TO REPAIR Professional wording is **evidence** of accuracy.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Verification	Checking whether a claim is supported by appropriate evidence
Citation	A reference that helps a reader locate a source
Unit	The agreed measurement scale attached to a quantity, such as centimetres
Corroboration	Additional evidence that supports a claim, with its origin checked

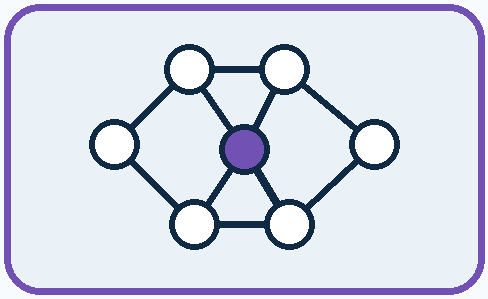
Follow a worked example

> Follow the worked trace for Verify Important Outputs, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a **claim** to observable **evidence**.



The case to investigate

CASE A generated paragraph confidently **claim**s that an animal lives in a region; the wording is not **evidence** that the **claim** is true.

Step 1

STEP 1 — OBSERVE specific **claim** extracted from an **output**

Step 2

STEP 2 — TRACE check **independent** suitable **source**s and reconcile differences

Step 3

STEP 3 — CONCLUDE supported, corrected, uncertain, or removed **claim**

Model answer

MODEL REASONING Check the exact **claim**, not the whole paragraph. Use **source**s with direct relevant **evidence**, note publication context, and explain whether they **independent**ly agree.

If **evidence** is weak or conflicting, revise or remove the **claim**.

A claim to avoid

DO NOT OVER**CLAIM** Two search results are not two **independent source**s when both copy the same page.

What new **evidence** could change this answer?

A closer explanation

Quantity	Calculation in centimetres	Equivalent in metres
Starting length	4 × 200 cm = 800 cm	8 m
Length used	12 × 50 cm = 600 cm	6 m
Length remaining	800 cm − 600 cm = 200 cm	2 m

Worked explanation · Verify Important Outputs

Convert to one length **unit** before subtracting. Two metres is 200 centimetres, so four rolls give 800 centimetres. The twelve pieces use 600 centimetres, leaving 200 centimetres, or 2 metres.

Subtracting the bare numbers 8 – 600 would mix metres with centimetres and would not describe the remaining length correctly.

Worked explanation · Verify Important Outputs

Use a second route as a check: 50 centimetres is half a metre, so twelve pieces use six metres. Eight minus six leaves two metres.

A reasonableness check also asks whether the used length is at most the starting length and whether starting length equals used plus remaining.

These checks can reveal mistakes; they do not replace checking the original **input**s and assumptions.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Verify Important Outputs

A fictional craft sheet calculates ribbon left after making bookmarks.

The facts we will use

Supplied facts

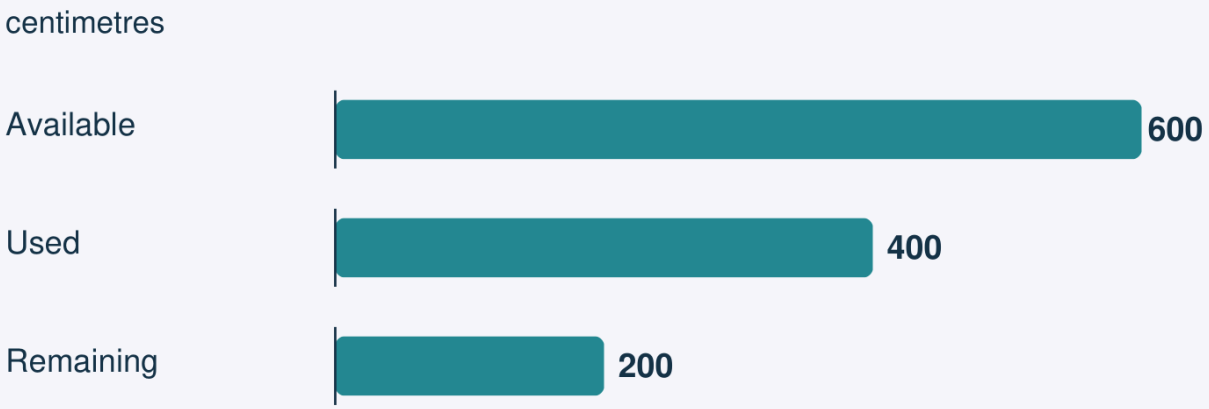
Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

Supplied facts

No other ribbon loss is assumed; no competition result is recorded.

AI ACADEMY / AI-1/27

Ribbon quantities in one consistent unit



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/27

Verify Important Outputs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Available ribbon

4 x 150=600 cm.



2. Ribbon used

8 x 50=400 cm.



3. Remainder

600-400=200 cm=2 m.



4. Audit draft

"200 m remains and everyone wins" fails unit and evidence checks.

CHANGE ONE THING

Make ten bookmarks instead of eight.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Available ribbon	4×150=600 cm.
2. Ribbon used	8×50=400 cm.
3. Remainder	600−400=200 cm=2 m.
4. Audit draft	“200 m remains and everyone wins” fails unit and evidence checks.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

All arithmetic first uses centimetres. Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**.

What this example does not establish

Keep the stated limit

Real cutting losses or inaccurate roll lengths would change this toy calculation; the assumption must be declared.

Keep the stated limit

Read the labelled picture: Correct the draft and identify which part needs another **source** rather than a calculation.

Keep the stated limit

Change one thing: Make ten bookmarks instead of eight. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Verify Important Outputs claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Generated sentence	fluent output	claim, not evidence
Zoo care page	direct animal information	possible source
Two copied blogs	same origin	not independent
Conflicting references	evidence disagrees	report uncertainty

Repair two tempting claims

Challenge claim A

CLAIM A Professional wording is **evidence** of accuracy.

Repair:

Challenge claim B

CLAIM B The first two agreeing links settle every **claim**.

Repair:

A frame for explaining

STRONG ANSWER FRAME I **claim** _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

A useful correction names the original **claim**, relevant **source**, calculation with **unit**s and revised conclusion. For example: R lists four 2-metre rolls and twelve 50-centimetre pieces.

The pieces use 600 centimetres, or 6 metres; the paper model leaves 2 metres when no cutting loss is included. This preserves the condition on which the result depends.

Check the claim against the facts

If the real activity loses ribbon at each cut, the no-loss result cannot simply be reused. Obtain that missing information or label the answer as a model with an explicit assumption.

Similarly, if you cannot access a **source**, mark the **claim** unresolved and identify what **evidence** would settle it. Repeating a confident answer is not a new measurement.

In a changed paper model, each of the twelve pieces uses 50 cm and an extra 2 cm is lost for each piece. Calculate the total loss and the remaining ribbon from the same four rolls. State which assumption changed

STUDENT LESSON · 8 OF 18

Practise together

Practise Verify Important Outputs with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed **independent**ly later.



Explain how the picture story demonstrates this idea: Fluent AI **output** is not **evidence**; important **claim**s need **independent** support.

Use a fresh example to explain **claim**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: The first two agreeing links settle every **claim**.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

The **claim** “600 metres used” conflicts with the stated **input**s and calculation. The **claim** “everyone will enjoy it” is not supported by R or by any supplied enjoyment **evidence**. That does not establish that nobody will enjoy it.

State “not verified from these records” rather than changing an unsupported positive **claim** into an unsupported negative **claim**.

> **Audit the draft from section one. For each of its five claims, state supported, contradicted or unresolved under the supplied evidence. Correct the used-length claim and explain why the correct remaining-length claim still needs its own calculation**

Suppose three posters repeat the same number and each copied it from one unverified draft. That is three copies of one **claim**, not three **independent** measurements. Follow references back to their origin.

A separate check of R and the arithmetic provides more useful **evidence** than counting copies that repeat the same mistake.

Practise with less help: Repair a unit error

Use workbook W2 for the connected example “Verifying **claim**s and quantities”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Verifying **claim**s and quantities. Proposed explanation: “A **citation** proves every nearby **claim**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Correct the draft and identify which part needs another **source** rather than a calculation.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named **independent** case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

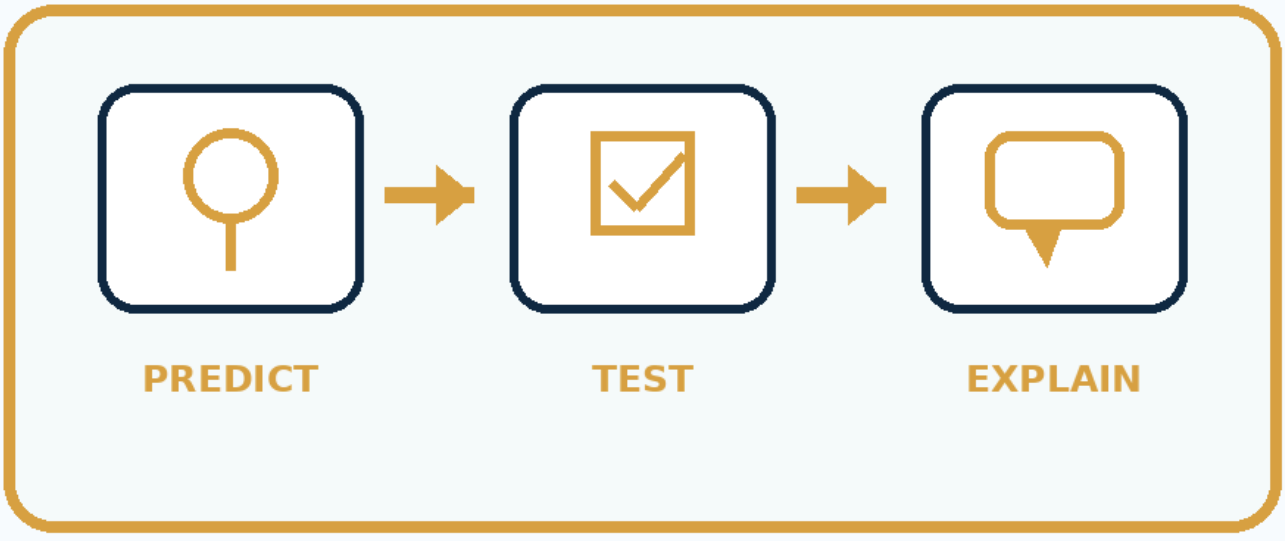
Run the investigation

> Investigate Verify Important Outputs. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Two-Source Claim Check

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Use a two-**source** checklist on a generated animal fact.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

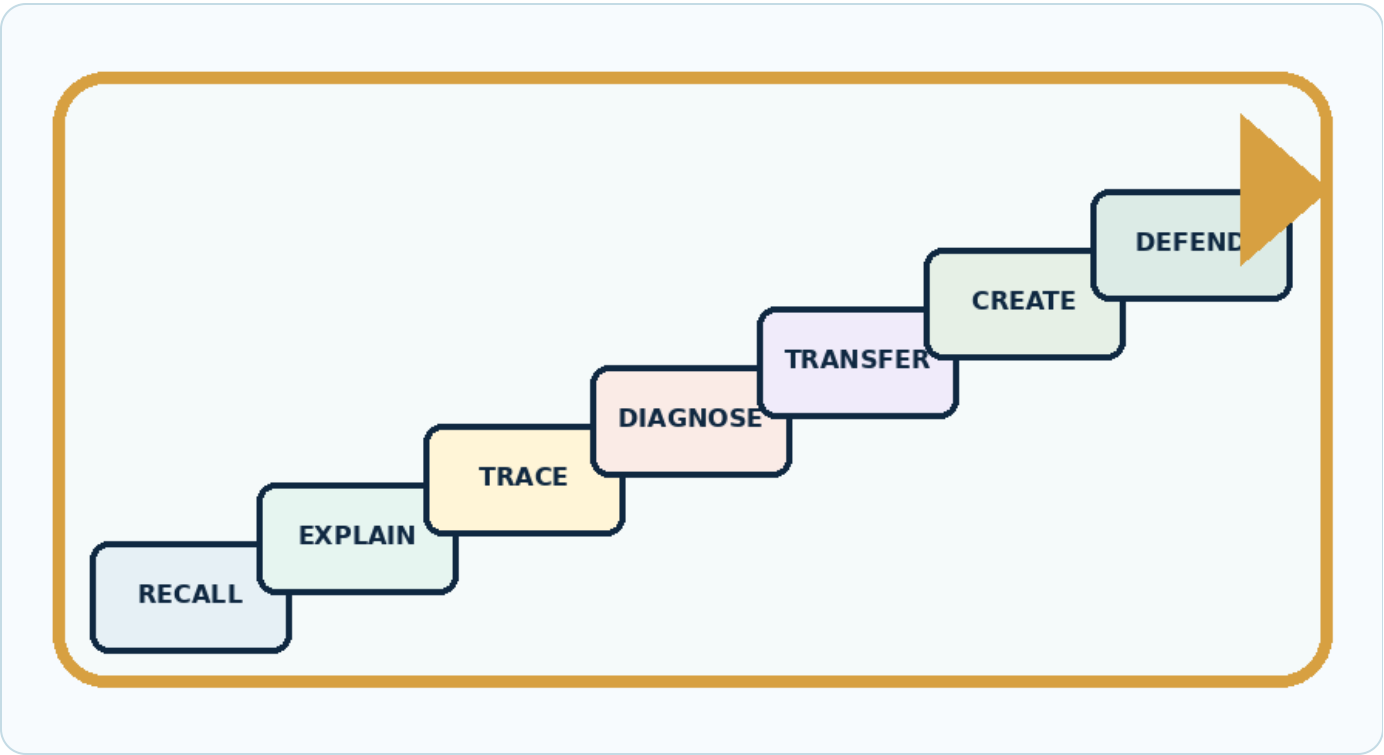
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Verify Important Outputs. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **claim**.

EXPLAIN Fluent AI **output** is not **evidence**; important **claim**s need **independent** support.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: Professional wording is **evidence** of accuracy.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a **claim**, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case **independent**ly.

STUDENT LESSON · 11 OF 18

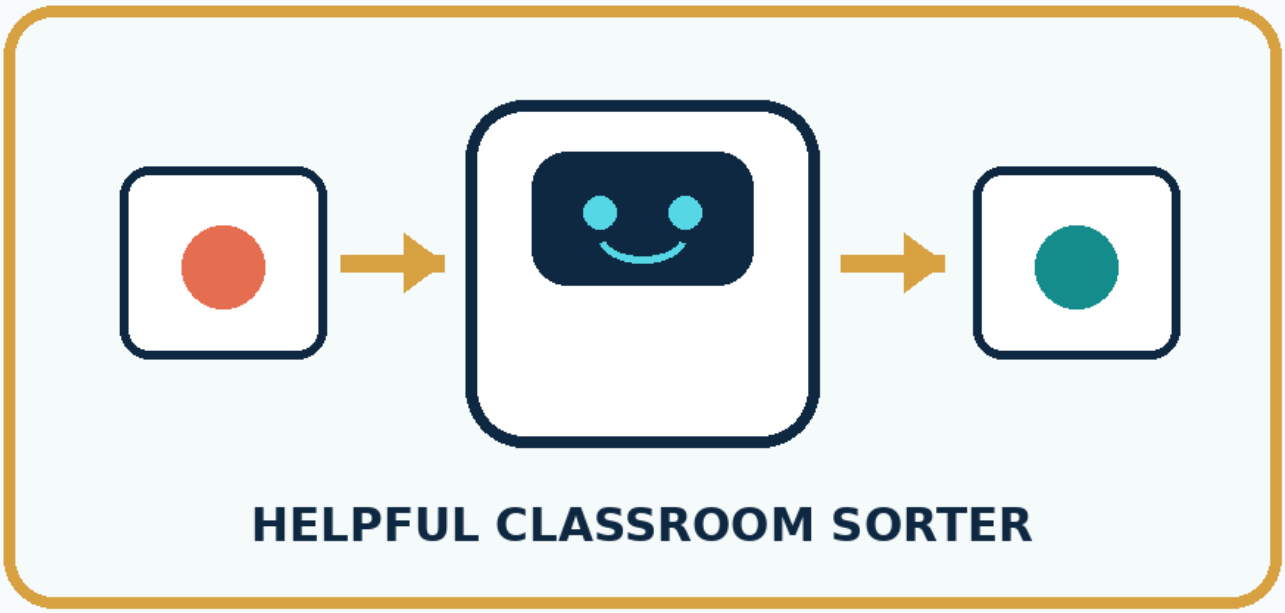
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Require human confirmation for important sorter **output**s and attach the **evidence** record used to approve each rule or label.

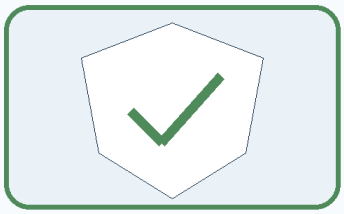
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Two search results are not two **independent source**s when both copy the same page.

My project decision:

Evidence for your project

Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Extract a generated animal **claim**, verify it with two **independent**

sources, and record support, correction, or uncertainty.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent evidence:

Check your explanation

SELF-CHECK Check the exact **claim**, not the whole paragraph. Use **source**s with direct relevant **evidence**, note publication context, and explain whether they **independent**ly agree.

If **evidence** is weak or conflicting, revise or remove the **claim**.

My level: ☐ Not yet ☐ With one hint ☐ **Independent**ly ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Humans Set Boundaries.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Verify Important Outputs case for supported practice. Repeated

> examples do not become fresh assessments.

Week 27 Worked case

Use the supplied details to practise the weekly concept before attempting the **independent** question. This case extends the illustrated lesson.

The situation

An answer says six groups of four make 28.

The question

Verify **independent**ly.

Worked explanation

Six times four is 24. Draw six groups or add four six times. Confident wording does not change arithmetic.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Six groups of four make twenty-four

Evidence or stage	Value or result
Group 1	4

Evidence or stage	Value or result
Group 2	4
Group 3	4
Group 4	4
Group 5	4
Group 6	4

An answer says six groups of four make 28.

Verify **independent**ly.

Six times four is 24. Draw six groups or add four six times. Confident wording does not change arithmetic.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Audit a new model with mass units

Fictional craft card M states five sealed bags of modelling material at exactly 250 g per bag. Three models each use 300 g, with no other loss.

A draft says “M supplies 1250 g; the models use 900 g; 350 kg remains; every model will win a prize.” Check all four **claim**s, show the remaining mass in grams, and explain the status of the prize **claim**. Use 1 kg = 1000 g.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 27 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An assistant says the school library opens at 8, but today's official notice says 9. Which **source** should guide today's visit? Explain the check.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Require human confirmation for important sorter **output**s and attach the **evidence** record used to approve each rule or label.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the **independent** answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A generated report says 18 of 24 tests passed, so success was **80%**. Verify the **claim** in two ways and write a corrected sentence.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the **independent** case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Verify Important Outputs, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional craft sheet calculates ribbon left after making bookmarks.

By the end, I can	How I will show it
Separate an answer into claims and connect each claim to relevant evidence	Explain the deciding step, show a trace or test, and state one limit.
Recalculate quantities with consistent units and a reasonableness check	Explain the deciding step, show a trace or test, and state one limit.
Report supported, contradicted and unresolved claims without inventing missing evidence	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try **independent**ly → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Verify Important Outputs without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Separate an answer into **claim**s and connect each **claim** to relevant **evidence**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved **independent** assessment for a separate first attempt before teacher feedback.

A fictional craft sheet calculates ribbon left after making bookmarks.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? “200 m remains and everyone wins” fails **unit** and **evidence** checks.

1. Audit draft

2. Remainder

3. Ribbon used

4. Available ribbon

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Audit draft “200 m remains and everyone wins” fails **unit** and **evidence** checks. All arithmetic first uses centimetres.

Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Make ten bookmarks instead of eight. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. 100 cm, or 1 m, remains.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: 100 cm, or 1 m, remains.

> 600−10×50=100 cm.

Repair a missing building block

Review Prompts Are Design Instructions (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Review Good Data, Bad Data (ai-1/21). Goal: Check records against a stated format, **unit** and meaning before using them. Short practice: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

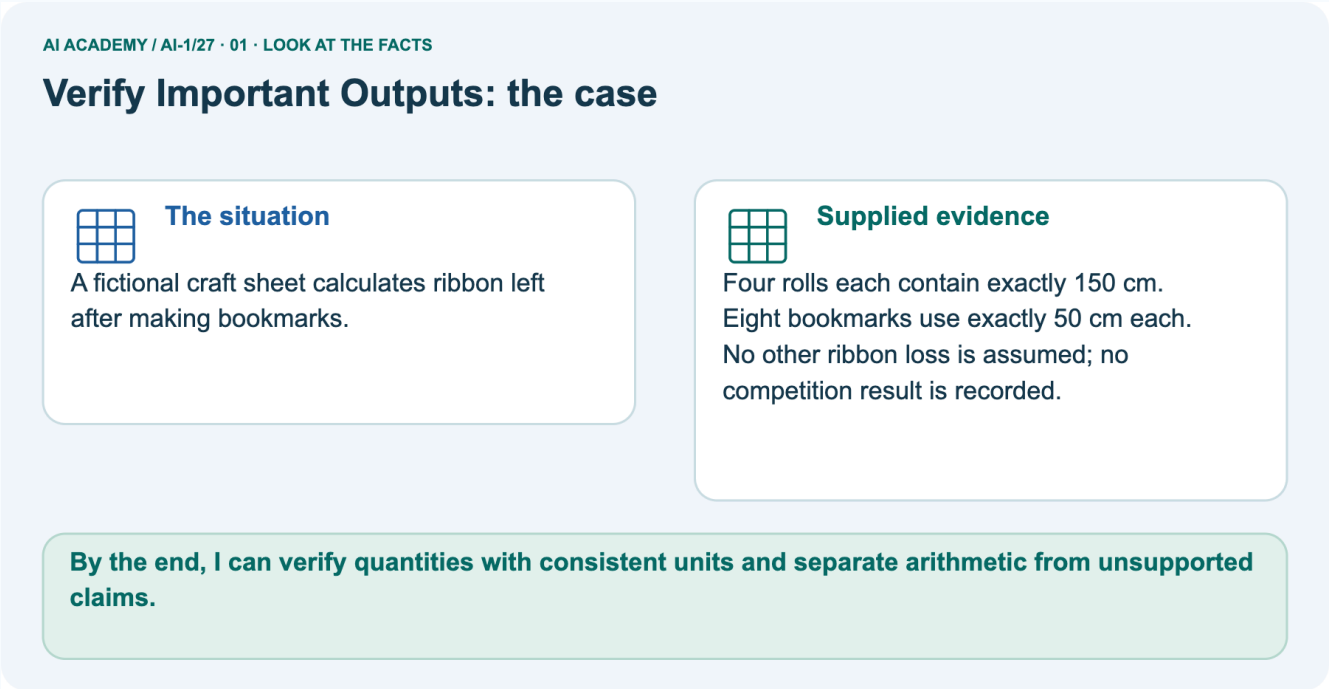
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Verify Important Outputs graphic using its labels, deciding evidence and relevant limitation.

01. Verify Important Outputs: the case



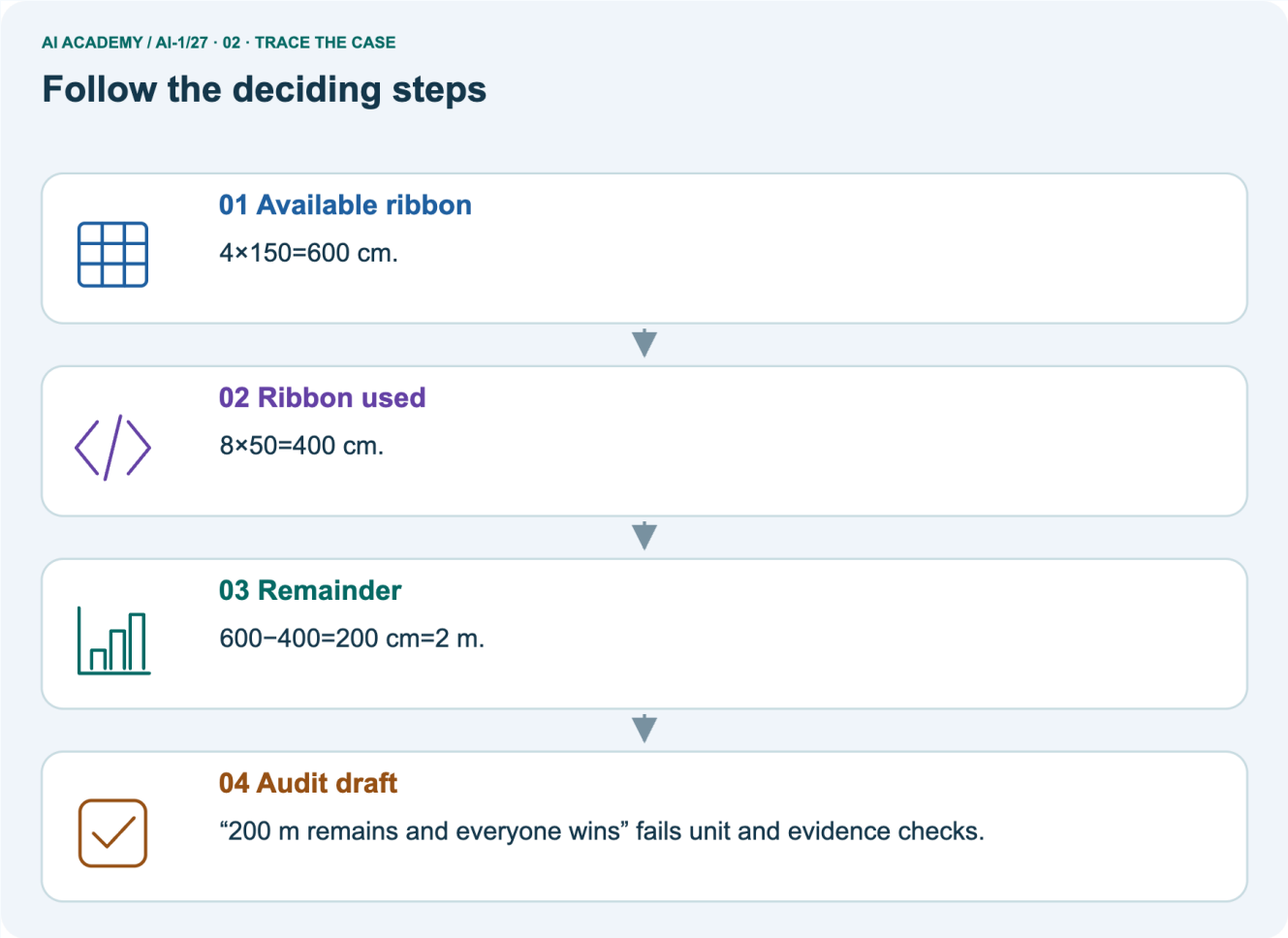
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft sheet calculates ribbon left after making bookmarks. Supplied facts: Four rolls each contain exactly 150 cm. Eight bookmarks use exactly 50 cm each.

No other ribbon loss is assumed; no competition result is recorded.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Available ribbon: 4×150=600 cm. → Ribbon used: 8×50=400 cm. → Remainder: 600−400=200 cm=2 m. → Audit draft: “200 m remains and everyone wins” fails **unit** and **evidence** checks.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Ribbon quantities in one consistent unit

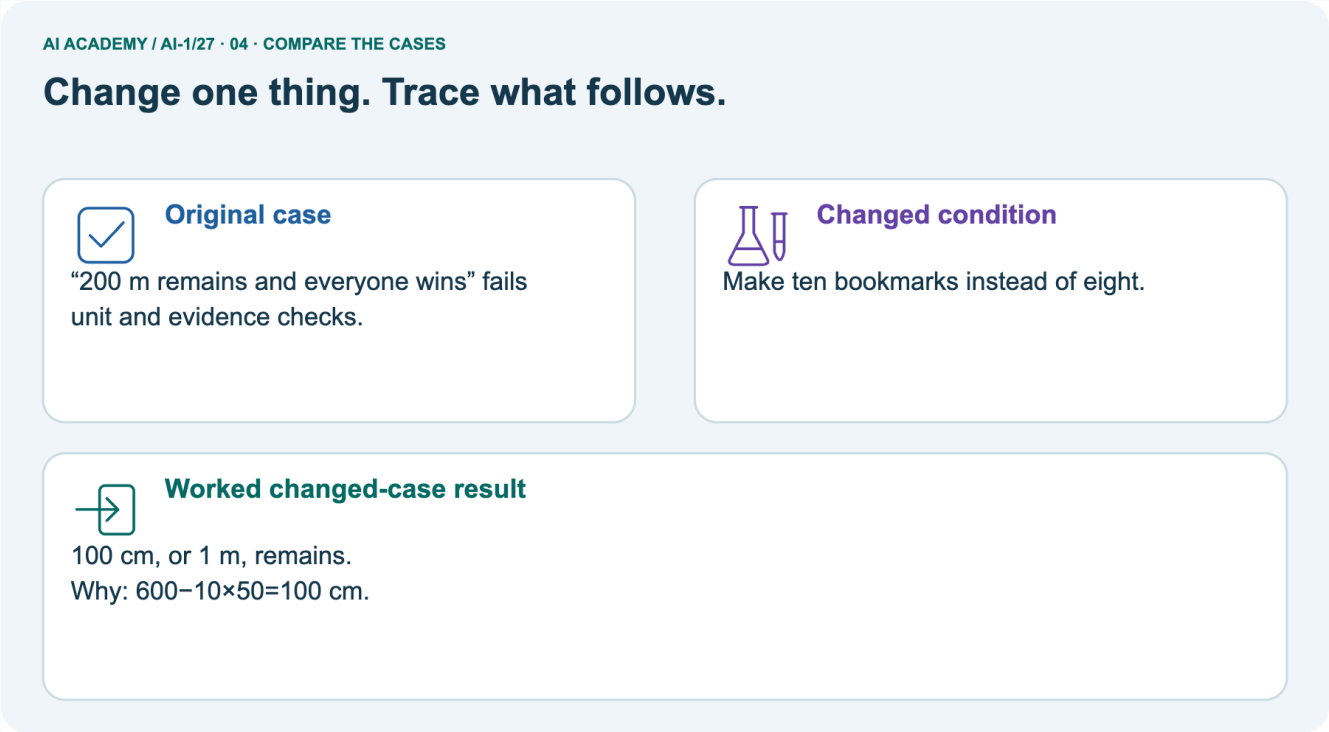


Follow the labelled picture with the matching explanation. Open the vector diagram

Ribbon quantities in one consistent **unit**. **Unit**s: centimetres. Available: 600. Used: 400. Remaining: 200.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

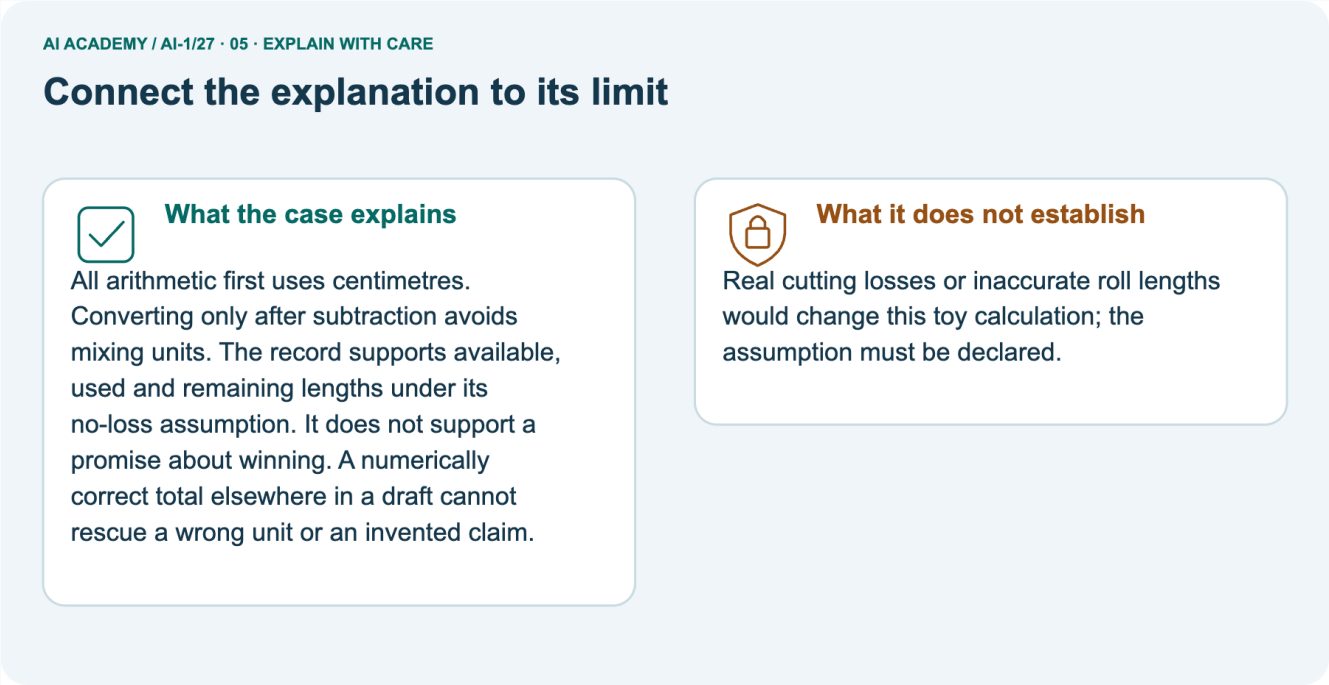


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: “200 m remains and everyone wins” fails **unit** and **evidence** checks. Changed condition: Make ten bookmarks instead of eight. Expected result: 100 cm, or 1 m, remains. Reason: 600−10×50=100 cm.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



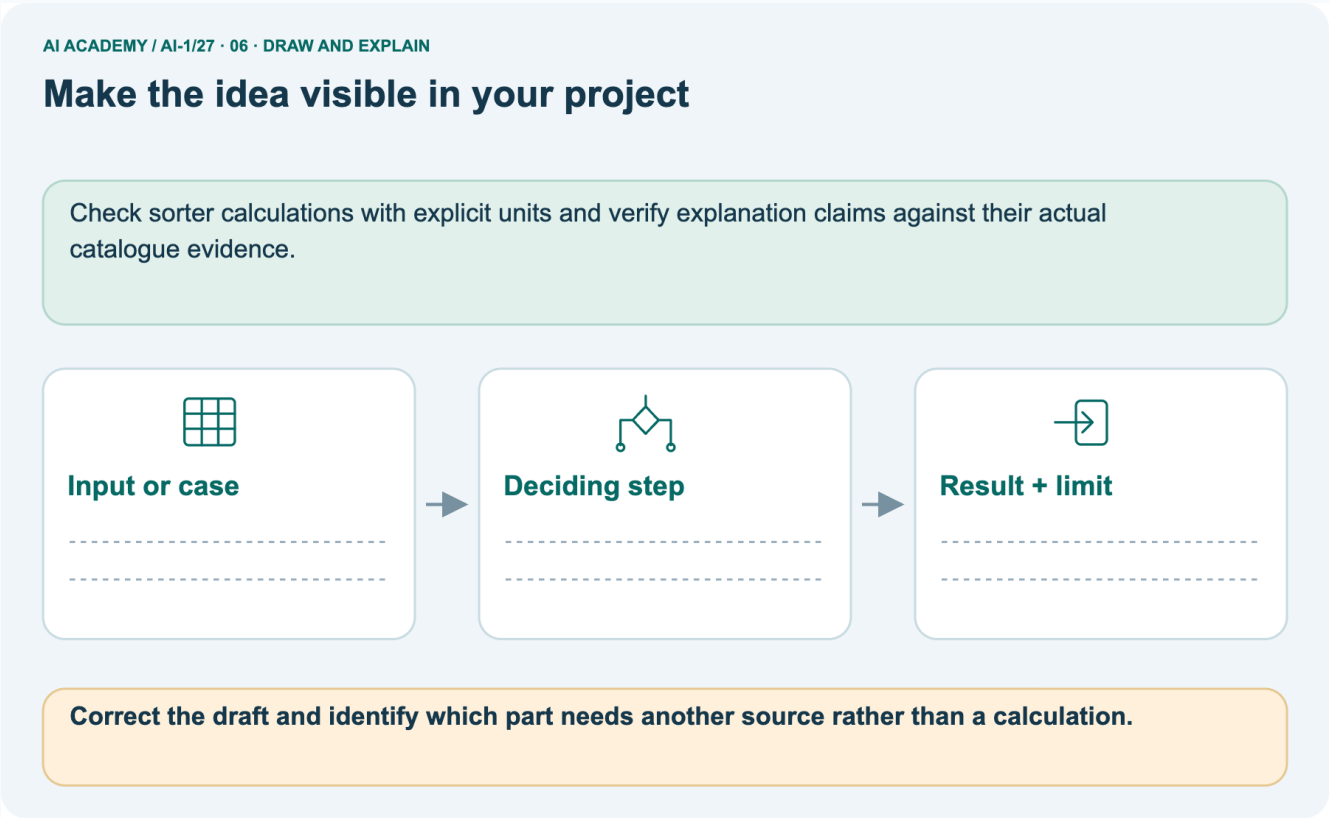
Follow the labelled picture with the matching explanation. Open the vector diagram

All arithmetic first uses centimetres. Converting only after subtraction avoids mixing **unit**s. The record supports available, used and remaining lengths under its no-loss assumption. It does not support a promise about winning.

A numerically correct total elsewhere in a draft cannot rescue a wrong **unit** or an invented **claim**. Limit: Real cutting losses or inaccurate roll lengths would change this toy calculation; the assumption must be declared.

> Explain the case, then point to a limit on the **claim**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Check sorter calculations with explicit **unit**s and verify explanation **claim**s against their actual catalogue **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Correct the draft and identify which part needs another **source** rather than a calculation.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.